of Slim's. This reflects the fact that as Jim gets farther out along the strut, his circular speed increases toward that of Slim.

We conclude that to observers on the spinning ride, such as Slim and Jim, the rate of passage of time depends upon their precise position—in this case, their distance from the center of the ride. This is an illustration of what we mean by warped time: Time is warped if its rate of passage differs from one location to another. And of particular importance to our present discussion, Jim will also notice something else as he crawls out along the strut. He will feel an increasingly strong outward pull because not only does speed increase, but his acceleration increases as well, the farther he is from the spinning ride's center. On the Tornado ride, then, we see that greater acceleration is tied up with slower clocks—that is, greater acceleration results in a more significant warping of time.

These observations took Einstein to the final leap. Since he had already shown gravity and accelerated motion to be effectively indistinguishable, and since he now had shown that accelerated motion is associated with the warping of space and time, he made the following proposal for the innards of the "black box" of gravity—the mechanism by which gravity operates. Gravity, according to Einstein, is the warping of space and time. Let's see what this means.

The Basics of General Relativity

To get a feel for this new view of gravity, let's consider the prototypical situation of a planet, such as the earth, revolving around a star, such as the sun. In Newtonian gravity the sun keeps the earth in orbit with an unidentified gravitational "tether" that somehow instantaneously reaches out across vast distances of space and grabs hold of the earth (and, similarly, the earth reaches out and grabs hold of the sun). Einstein provided a new conception of what actually happens. It will aid in our discussion of Einstein's approach to have a concrete visual model of spacetime that we can conveniently manipulate. To do so, we will simplify things in two ways. First, for the moment, we will ignore time and focus solely on a visual model of space. We will reincorporate time in our discussion shortly. Second, in order to allow us to draw and manipulate visual images on the pages of this book, we will often refer to a two-dimensional analog of three-dimensional space. Most of the insight we gain from thinking in terms